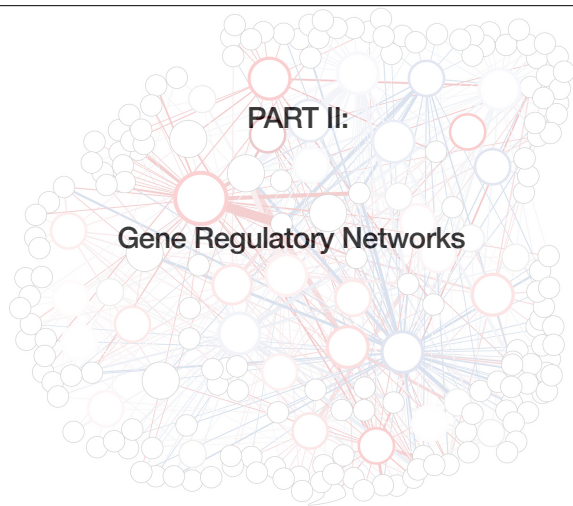
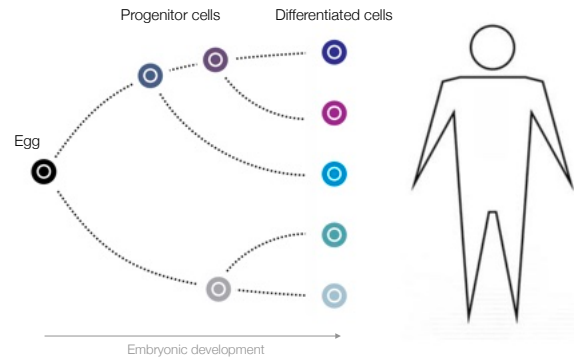
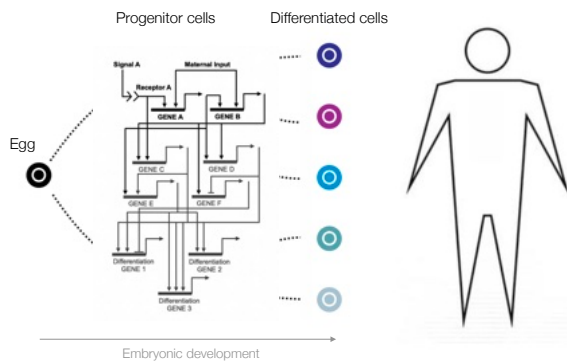




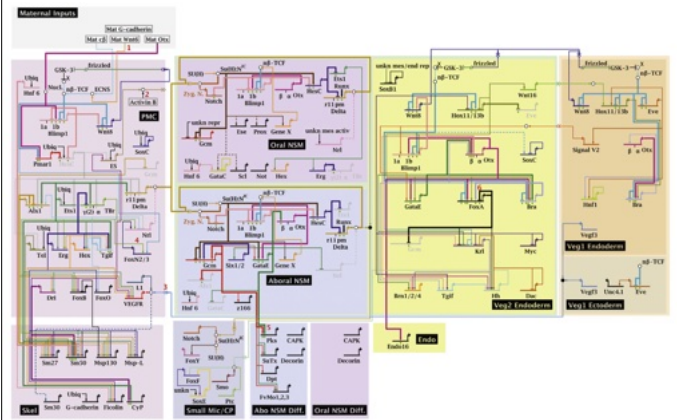
Cell identity is governed by gene regulatory networks (GRNs)



Cell identity is governed by gene regulatory networks (GRNs)



“Development of the body plan is controlled by large networks of regulatory genes”



The Britten-Davidson Model (1969): A conceptual framework for Gene Regulatory Networks



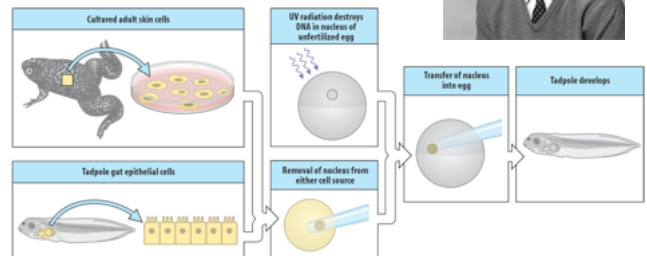
Cell differentiation is based almost certainly on the regulation of gene activity, so that for each state of differentiation a certain set of genes is active in transcription and other genes are inactive.

PREMISES:

1. Cells contain identical genomes
2. Different cells express different sets of genes

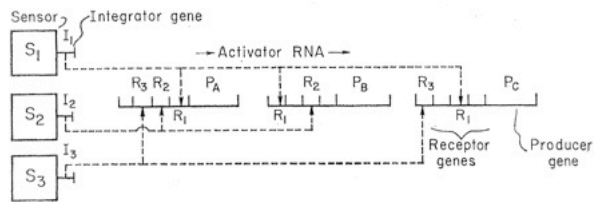
Principle of Genomic Equivalence Sir John Gurdon, 1962

Nobel Prize 2012



Adult cells contain all the information necessary to generate an adult organism. Hence, how this information is interpreted must define cell identity.

The Britten-Davidson Model (1969) A conceptual/logical Framework for Gene Regulatory Networks



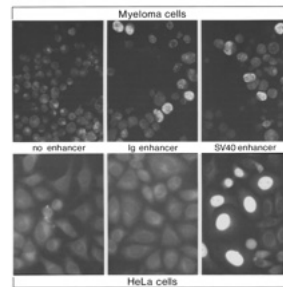
- Offered a logical structure for **regulation of gene activity**
- Proposed a mechanism for phenotypic evolution

Open Question: What is the specific structure of the network?
(experimental challenge)

Tissue-specific gene expression is controlled by enhancers

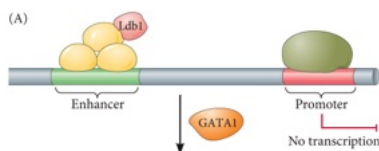
A Lymphocyte-Specific Cellular Enhancer Is Located Downstream of the Joining Region in Immunoglobulin Heavy Chain Genes
Julian Benayahu, Laura Olson, and Walter Schaffner

Testing enhancers using mouse transgenesis:



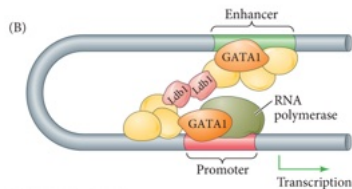
E11.5

Enhancers are regulated by transcription factors



Enhancers: cis-regulatory elements

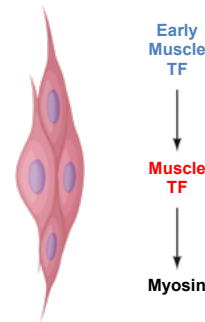
TFs: trans-regulators



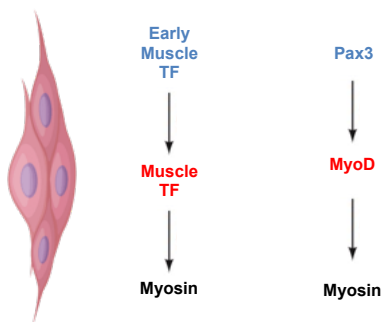
While multiple proteins are necessary for gene expression **only TFs can read DNA sequence.**

DEVELOPMENTAL BIOLOGY 11e, Figure 3.7
© 2016 Sinauer Associates, Inc.

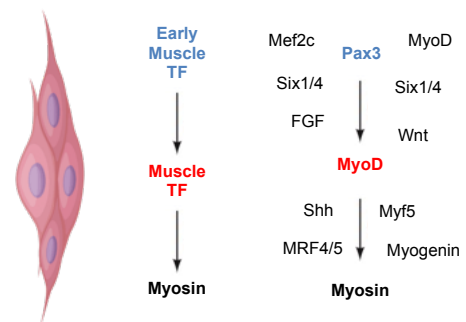
Regulatory programs controlling cell identity are complex



Regulatory programs controlling cell identity are complex

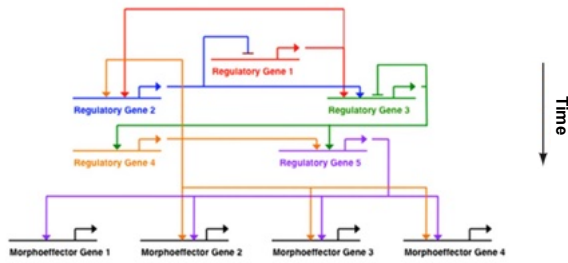


Regulatory programs controlling cell identity are complex



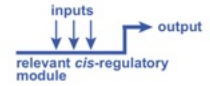
There is no "master regulator" of cell identity

Cell identity is controlled by complex gene regulatory networks

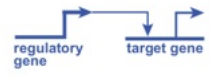


Components of topological GRN models as portrayed in BioTapestry

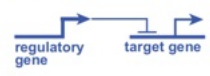
Node, Inputs, Output: a regulatory gene and its relevant *cis*-regulatory module(s), in which the inputs are the transcription factors encoded elsewhere in the GRN that bind to the *cis*-regulatory module(s) and the output is the transcription factor encoded by the gene at this node.



Positive regulatory linkage: interaction between two genes such that an upstream regulatory gene provides an activating input to a downstream target gene.



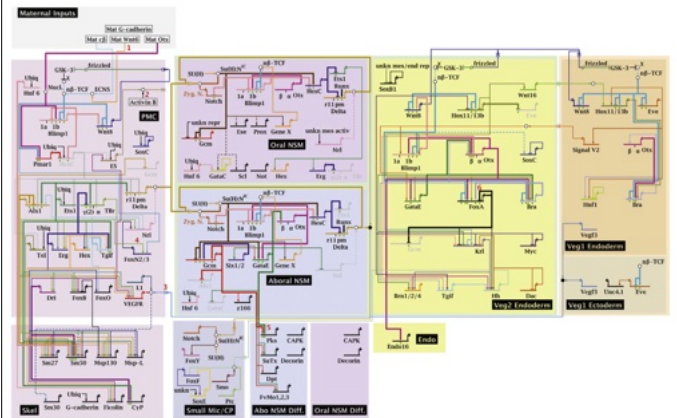
Negative regulatory linkage: interaction between two genes such that an upstream regulatory gene provides a repressive input to a downstream target gene.



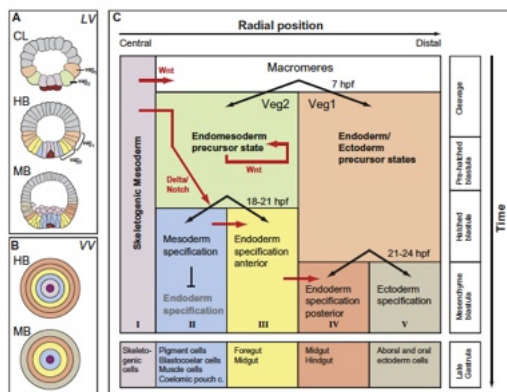
Principles of gene regulatory networks:

1. GRNs are MODULAR
2. GRNs are HIERARCHICAL
3. GRNs are based on CAUSAL relationships

Regulatory networks are modular



Regulatory networks are modular: composed of "regulatory modules": SPACE and TIME

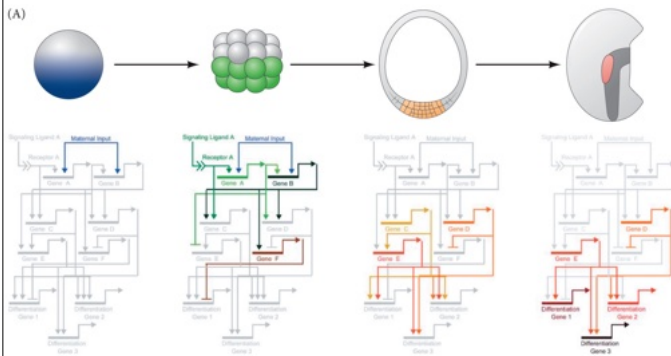


Regulatory networks are modular: composed of "subcircuits": FUNCTION

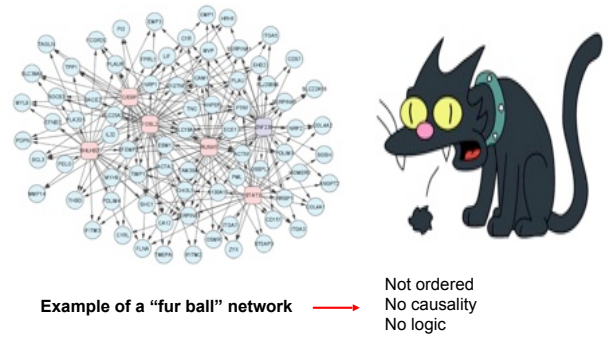


Fig. 3. Distribution of subcircuits in the endomesoderm GRN. Subcircuits of each type identified in the endomesoderm GRN model are color-coded as follows: pink, double-negative gate; dark blue, coherent feedforward subcircuit; light blue, community effect subcircuit; yellow, positive feedback subcircuit; green, signaling interaction; red, toggle switch circuitry; and brown, mutual-repression subcircuit. For recent updates of the endomesoderm GRN model, see grns.bio.tapestry.org/fig3/endomesoderm/.

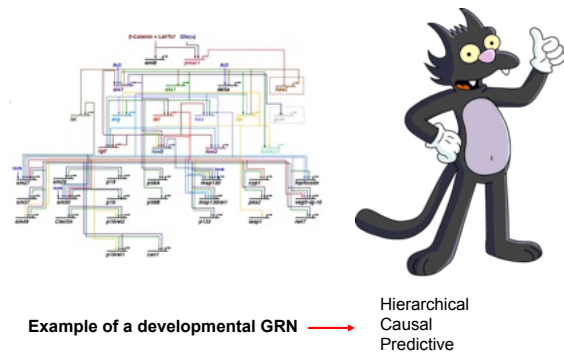
Gene regulatory networks are hierarchical and drive progressive specification



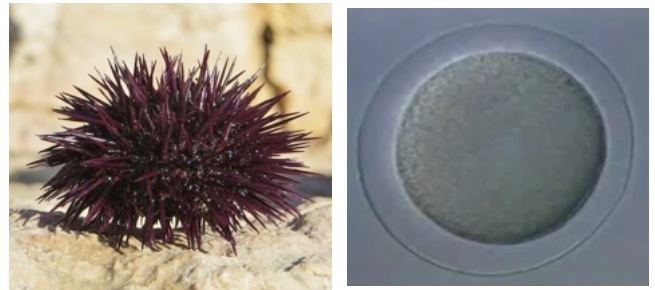
GRNs are composed of causal interactions



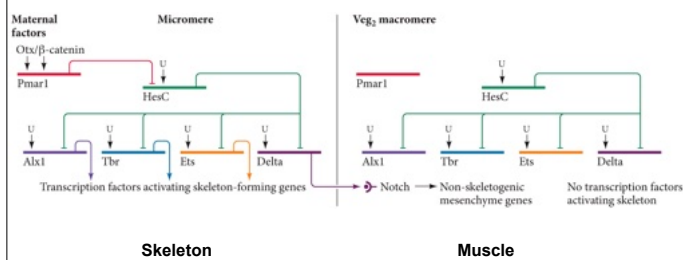
GRNs are composed of causal interactions



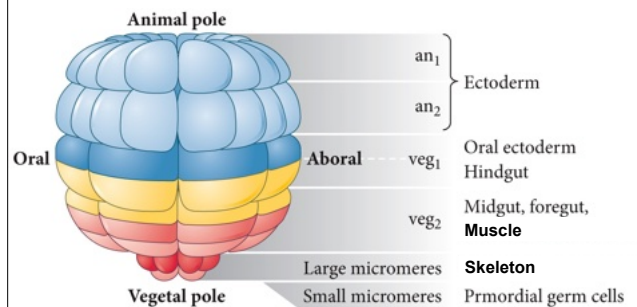
Fate determination in the sea urchin mesoderm



Gene regulatory circuit for skeletogenic specification



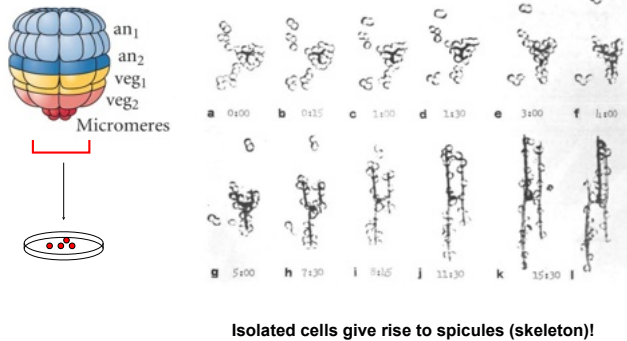
Fate map and cell lineage of sea urchin blastula



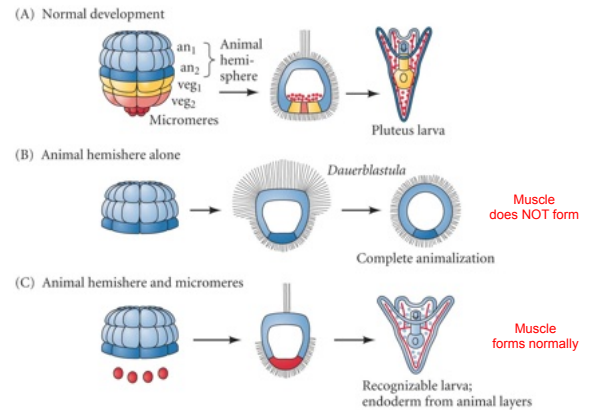
What is the gene regulatory circuit that controls cell fate specification of mesenchymal cells?

Micromere (muscle) vs Veg2 macromere (skeleton) mesoderm

Cell fates of skeletogenic micromeres are autonomously specified (isolation experiments by Okazaki, 1975)



Micromeres have the ability to induce presumptive ectodermal cells to acquire other fates



Conclusions

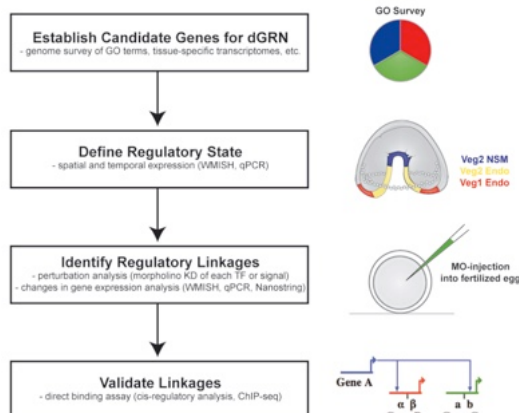
Fate determination of sea urchin mesoderm cells depend upon:

- Autonomous specification of micromeres (skeleton)
- Conditional specification of Veg2 macromeres (muscle), by the micromeres themselves!

1) What is the molecular basis for the cell-autonomous determination of the micromeres?

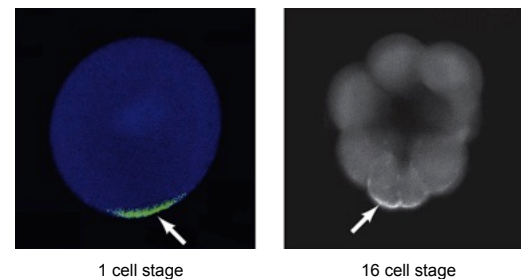
2) What are the inducing signals produced by the micromeres?

How do we assemble gene regulatory networks?



Studies of gastrulation led to insight into determination of micromeres

The protein *Dishevelled* (DSH) accumulates in the vegetal pole of sea urchin embryos



Dishevelled is a component of the Wnt signaling pathway

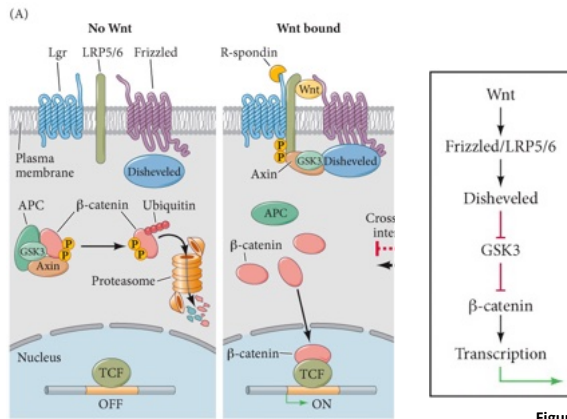
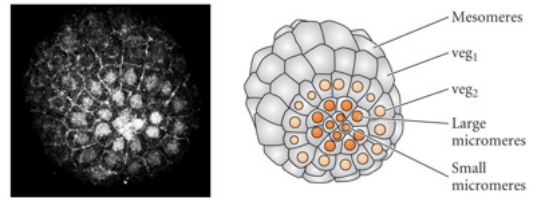


Figure 4.34

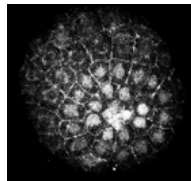
Studies of gastrulation led to insight into determination of micromeres



β -catenin is present in the nuclei of vegetal cells!

Hypothesis: β -catenin is responsible for specifying the fate of the micromeres

How to test this?



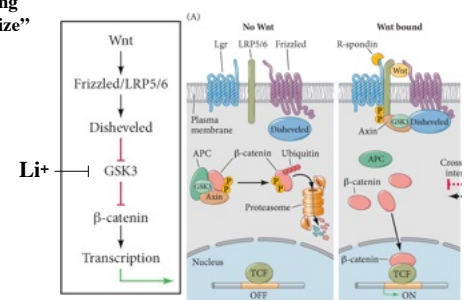
Test 1: What happens if β -catenin is present in nuclei of non-micromere cells? (sufficiency)

Test 2: What happens when β -catenin is prevented from entering the nucleus? (necessity)

How can one raise β -catenin levels in non-micromere cells?

Lithium chloride has long been known to “vegetalize” embryos.

GSK-3 is a target of Lithium chloride



Test 1: LiCl treatment of embryos

Result: increase in nuclear β -catenin correlates with vegetalization

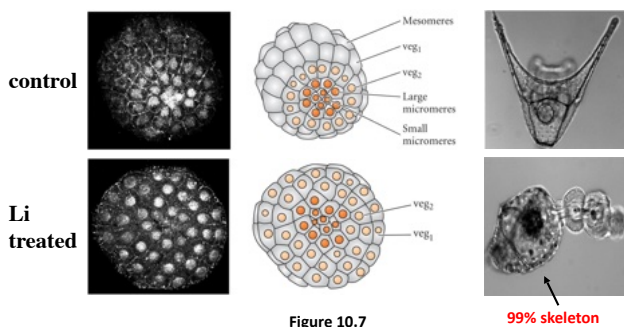
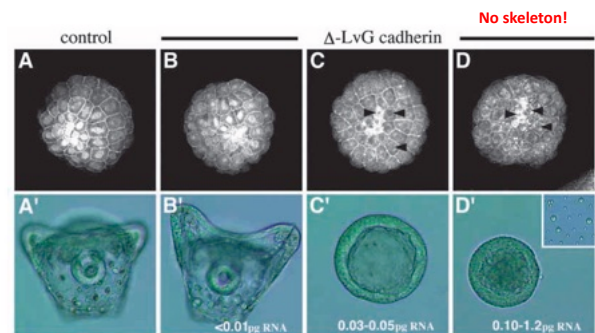


Figure 10.7

Nuclear β -catenin is sufficient to specify vegetal fates

Test 2: remove β -catenin from nucleus by injecting sequestering it with the intracellular domain of cadherin



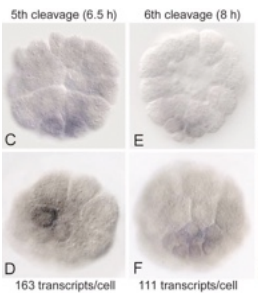
Logan et al. (1999) Development 126: 345

Nuclear β -catenin is necessary to specify vegetal fates

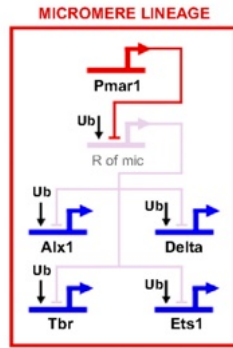
What does B-catenin do?



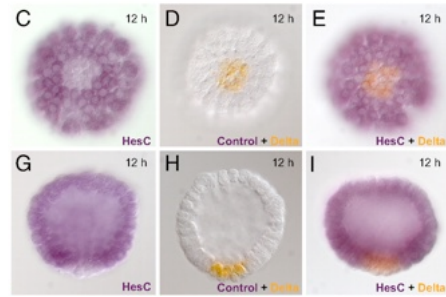
Activates Pmar1



Pmar1: transcriptional repressor

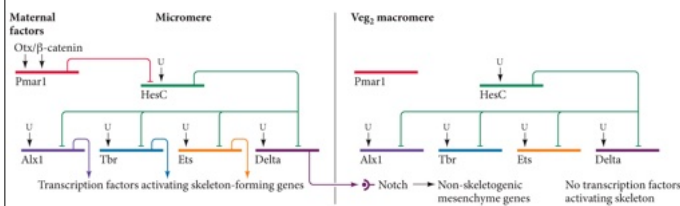


But what is PMAR repressing?



PMAR is repressing HesC, which is a ubiquitous repressor of skeletal genes!

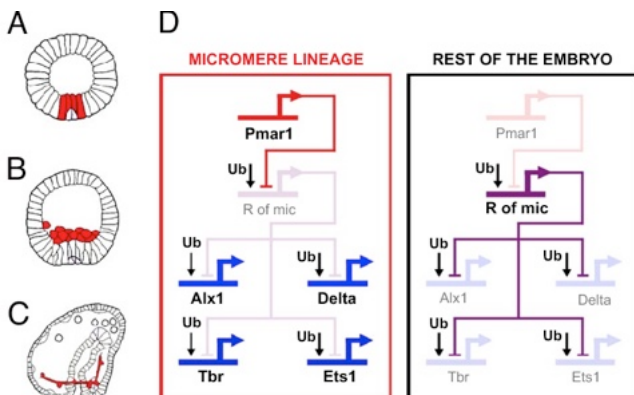
Gene regulatory circuit for micromere specification



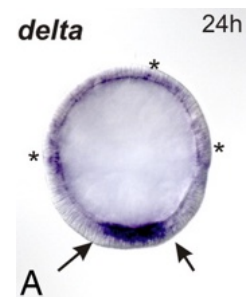
1). What is the molecular basis for the cell-autonomous determination of the micromeres?

Micromere fate is specified by nuclear β -catenin (skeletogenesis & ability to induce)

2) What are the inducing signals produced by the micromeres?



Large micromeres express Delta!



Notch signaling is an important pathway in cell-cell communication

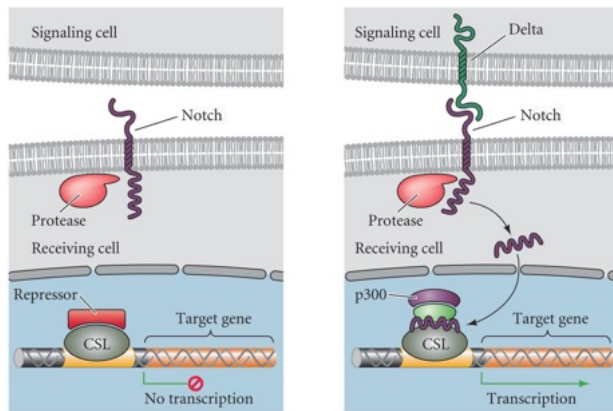


Figure 4.43

What about the conditional specification of the muscle cells?

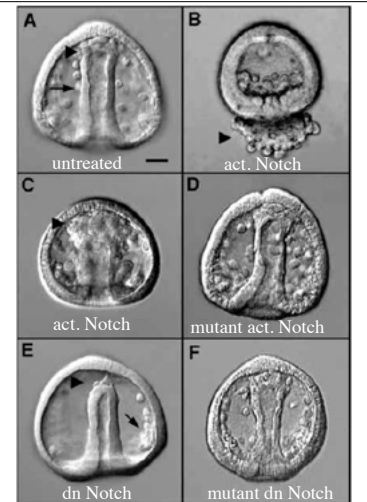
Test 1: Express activated Notch protein

Result: Excess muscle

Test 2: remove Notch activity by over-expression of dominant negative Notch protein

Result: Loss of muscle

From Sherwood & McClay (1999)
Development 126:1703



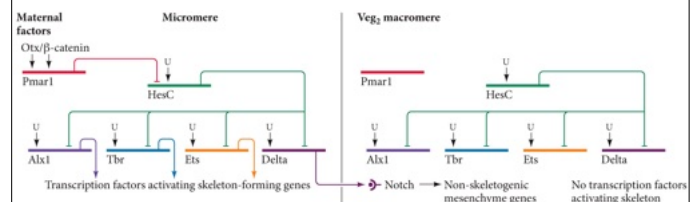
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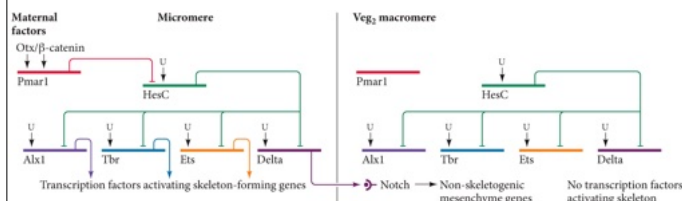
2) What are the inducing signals produced by the micromeres?

Micromeres induce muscle fate in the *Veg2* macromeres via the Notch/Delta signaling pathway

Gene regulatory circuit for micromere specification

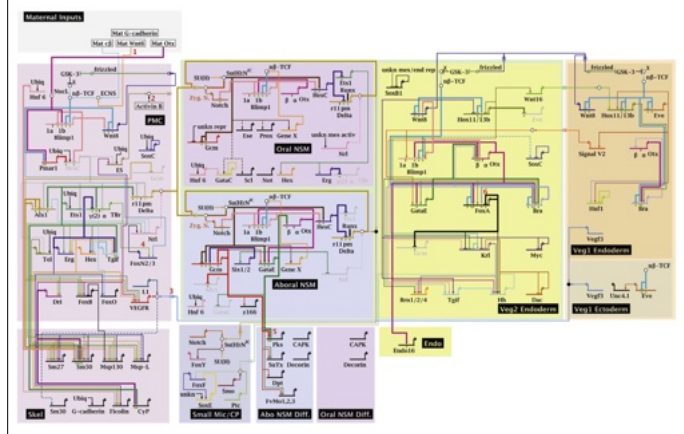


True or False?



- () Micromeres express high levels of HesC.
- () Knock down of Pmar1 or HesC in micromeres will result on the same phenotype.
- () Injection of dishevelled (DSH) in *Veg2* macromeres will result in the formation of ectopic skeleton.
- () *Veg2* macromeres cultured in isolation are able to form non-skeletogenic mesenchyme (SMCs).
- () Knocking down the HesC gene in isolated *Veg2* macromeres will cause them to give rise to skeletal cells.

Current State of the Sea Urchin GRN

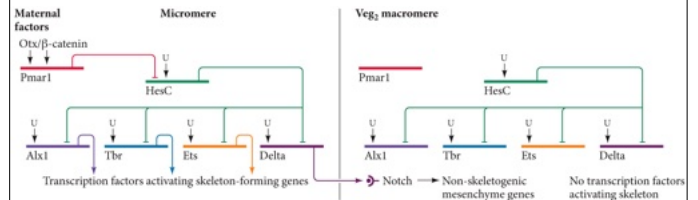


But are GRNs just overcomplicated genetic pathways?

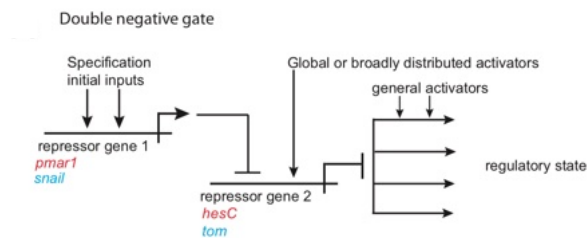
The answer is **NO**, for three reasons:

1. Network architecture determines regulatory logic.
2. Complete GRNs are predictive and allow for manipulation of cell fate
3. GRNs explain robustness, timing, and memory in development.

Take for instance the regulatory circuit underlying micromere specification:



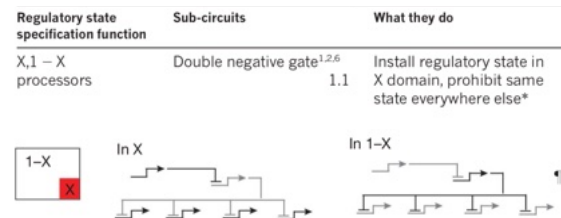
Regulatory logic of a double negative gate:



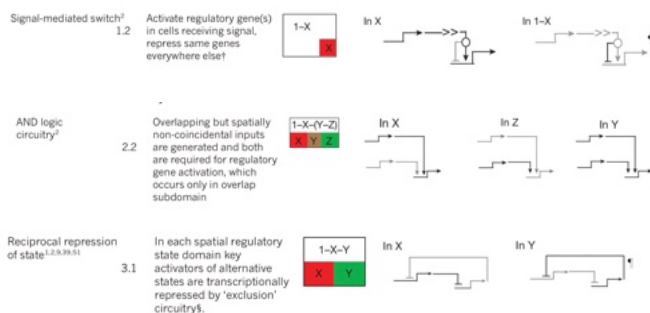
Regulatory genes that create initial regulatory state are controlled by **widely expressed repressor**, which is dominant over their positive inputs.

The gene encoding this repressor is itself specifically repressed in a local region (X) by another gene encoding a different repressor: hence target genes are ON in X, specifically repressed elsewhere.

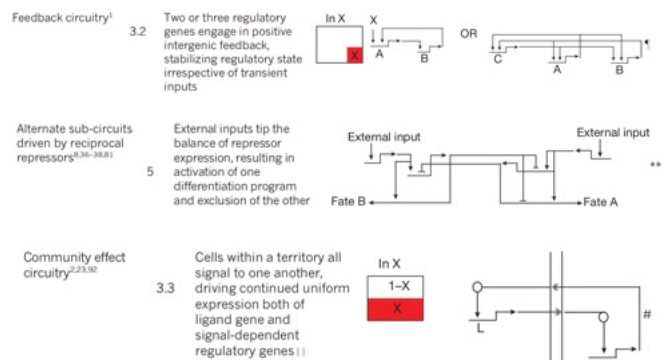
Regulatory logic of a double negative gate:



Other examples of sub-circuit logic in GRNs



Other examples of sub-circuit logic in GRNs



Sub-circuit logic is thoroughly employed in developmental GRNs



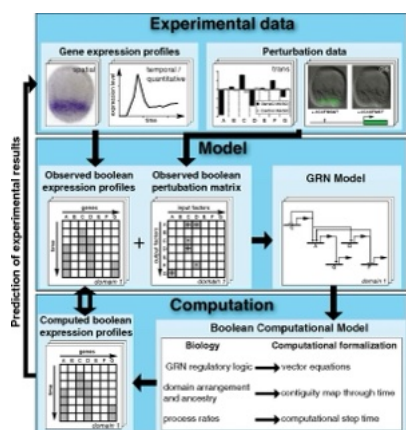
Fig. 3. Distribution of subcircuits in the endomesoderm GRN. Subcircuits of each type identified in the endomesoderm GRN model are color-coded as follows: pink, double-negative gate; dark blue, coherent feedforward subcircuit; light blue, community-effect subcircuit; yellow, positive feedback subcircuit; green, signaling interaction; red, toggle switch circuitry; and brown, mutual-repression subcircuit. For recent updates of the endomesoderm GRN model, see grns.loria.univ-lorraine.fr/grn-endomesoderm/.

But are GRNs just overcomplicated genetic pathways?

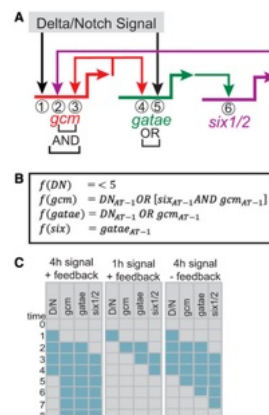
The answer is **NO**, for three reasons:

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3. GRNs explain robustness, timing, and memory in development.

Boolean modeling of the GRN



The model is predictive: control of non-skeletal genes in Veg 2 micromeres



Modeling GRNs with Ordinary Differential Equations

